YEAR 11
MATHS & SCIENCE

Completing the GCSE curriculum and converting knowledge into dependable examination performance

Government-source edition • Department for Education / GOV.UK only

Prepared for families supporting pupils through the final GCSE year.

1. How to use this guide

Government GCSE content is qualification-wide, not divided into a statutory Year 10 and Year 11 sequence. By Year 11, the task is therefore dual: complete any remaining specification content and integrate the whole course so knowledge can be retrieved and applied under examination conditions.

The Year 11 principle
Complete → connect → retrieve → apply → refine. Revision is not rereading; it is the deliberate conversion of the full curriculum into usable knowledge and exam performance.

2. GCSE Mathematics - the whole-course map

DOMAIN	WHAT MUST REMAIN ACTIVE
NUMBER	Structure and calculation; fractions, decimals and percentages; measures and accuracy; powers, roots, standard form and, for higher attainment, surds and fractional indices.
ALGEBRA	Notation, vocabulary and manipulation; formulae; equations and inequalities; sequences; graphs; coordinate geometry; functions and, at higher tier, more advanced algebraic reasoning.
RATIO, PROPORTION & RATES	Ratios and fractions; scale; direct and inverse proportion; percentage change; growth and decay; compound measures including speed, density, pressure and rates of change.
GEOMETRY & MEASURES	Properties and constructions; angle reasoning; transformations; mensuration; similarity and

	congruence; vectors; Pythagoras; trigonometry; circles; bearings and loci.
PROBABILITY	Single and combined events; relative frequency; sample spaces; tree diagrams and conditional probability at the appropriate level.
STATISTICS	Sampling; tables and charts; averages and spread; cumulative frequency and box plots where appropriate; scatter graphs; interpreting and comparing distributions.

3. Mathematics - Year 11 performance lens

FROM	TO
TOPIC-BY-TOPIC PRACTICE	Mixed papers and interleaved problems that force method selection.
KNOWING A METHOD	Recognising when, why and how to use it.
CORRECT WORKING	Efficient, legible working that earns method marks and supports checking.
REVISING STRENGTHS	Targeted repair of high-frequency weak prerequisites and error patterns.
ONE-OFF MOCK SCORES	Trend evidence across topics, papers and assessment objectives.
LAST-MINUTE CRAMMING	Spaced retrieval, timed practice and structured correction cycles.

4. GCSE Science - the whole-course map

DOMAIN	WHAT MUST REMAIN ACTIVE
BIOLOGY	Cell biology; organisation; infection and response; bioenergetics; homeostasis and response; inheritance, variation and evolution; ecology.
CHEMISTRY	Atomic structure and periodic table; bonding, structure and properties; quantitative chemistry; chemical changes; energy changes; rates and equilibrium; organic chemistry; chemical analysis; atmosphere; using resources.

PHYSICS	Energy; electricity; particle model of matter; atomic structure; forces; waves; magnetism and electromagnetism; space physics where included in the relevant qualification.
PRACTICAL & ENQUIRY	Required practical experiences and apparatus techniques; planning, variables, measurement, data handling, graphing, evaluation, uncertainty and drawing evidence-based conclusions.
MATHEMATICAL SKILLS	Use of equations, ratios, percentages, standard form, graphs, gradients, areas, proportional reasoning, rearrangement and numerical analysis in scientific contexts.

5. Science - Year 11 performance lens

FROM	TO
FACT RECALL	Precise recall plus application to unfamiliar scenarios.
PRACTICAL MEMORY	Understanding variables, controls, apparatus, uncertainty, graphs and evaluation.
EQUATION SUBSTITUTION	Selecting, rearranging, calculating, unit-converting and interpreting equations.
DESCRIPTIVE ANSWERS	Linked causal explanations using correct scientific vocabulary.
PAST PAPERS AS SCORES	Past papers as diagnostic evidence: classify every lost mark.
SEPARATE UNITS	Whole-course connections across biology, chemistry and physics.

6. A simple diagnostic system for every mock

ERROR TYPE	WHAT IT MEANS	RESPONSE
KNOWLEDGE	Fact, definition, equation or process not retrievable.	Retrieval practice and spaced re-testing.
METHOD	Student knew the topic but not the procedure.	Worked examples → guided practice → independent practice.

APPLICATION	Could not transfer knowledge to unfamiliar context.	More mixed/contextual questions with explanation of cues.
INTERPRETATION	Graph, data, wording or command word misread.	Target data-response and command-word practice.
EXECUTION	Arithmetic, units, signs, rounding, omitted working or timing.	Checklists, timed sets and deliberate accuracy routines.

7. Parent support plan

- Make the specification and school curriculum map the revision boundary - not random online topic lists.
- Use mock analysis to select the next week's work.
- Balance knowledge retrieval with exam questions; neither is sufficient alone.
- Keep mathematics and science calculation skills live every week.
- Preserve sleep, attendance and realistic routines: Year 11 gains come from consistency more than heroic bursts.

8. What to obtain from your child's school

- Final specification and tier entries for maths and science.
- Dates and content ranges for mocks and final internal assessments.
- Any remaining untaught specification content.
- Required-practical coverage and intervention plans.
- School revision resources and exam-board papers/question banks that align with the specification.

Official government sources used

The curriculum content in this guide is derived from Department for Education / GOV.UK statutory programmes of study and GCSE subject-content publications. The year-specific sequencing lens is labelled as an Inspire support framework because government documents generally specify content by key stage or qualification, not by individual secondary-school year.

Department for Education. National curriculum in England: mathematics programme of study - key stage 4.

Ref: DFE-00496-2014

Published within the national curriculum mathematics programmes of study.

https://www.gov.uk/government/publications/national-curriculum-in-england-mathematics-programmes-of-study

Department for Education. National curriculum in England: science programme of study - key stage 4.

Ref: DFE-00677-2014

Published 2 December 2014 within the national curriculum science programmes of study.

https://www.gov.uk/government/publications/national-curriculum-in-england-science-programmes-of-study

Department for Education. GCSE mathematics: subject content and assessment objectives.

Ref: DFE-00233-2013

Published 1 November 2013.

https://www.gov.uk/government/publications/gcse-mathematics-subject-content-and-assessment-objectives

Department for Education. GCSE subject content for combined science.

Ref: DFE-00351-2014

Published 9 April 2014; last updated 3 July 2015.

https://www.gov.uk/government/publications/gcse-combined-science

Department for Education. GCSE subject content for biology, chemistry and physics.

Ref: DFE-00352-2014

Published 9 April 2014; last updated 3 July 2015.

https://www.gov.uk/government/publications/gcse-single-science